

Design Document

Project Title: NoteShare

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Video Link: <https://youtu.be/bj2lG2XPpSY>

PowerBI Link:

<https://app.powerbi.com/groups/me/reports/3713411c-c0ad-419d-8b20-f531b45bd38e/73b728dececbd3d64ebf?experience=power-bi>

Purpose of the Database:

The purpose of this database is to manage information related to music, including artists, albums, songs, playlists, and users. It is designed to organize and store data about music tracks, albums, and artists, allowing users to create playlists, manage their music preferences, and discover new music. This database is what I would assume a Spotify or Apple Music type of app would have on the back end with added features such as knowing BPM.

Entities, Relationships, and Optimizations:

- Artist Table:
 - ArtistID (INT): Primary key for the artist.
 - ArtistName (VARCHAR(255)): Stores the name of the artist.
- Album Table:
 - AlbumID (INT): Primary key for the album.
 - AlbumName (VARCHAR(255)): Stores the name of the album.
 - ArtistID (INT): Foreign key referencing the Artist table.
 - ReleaseDate (DATE): Stores the release date of the album.
 - AlbumType (VARCHAR(50)): Indicates the type of the album (e.g., studio album, live album).
- Song Table:

- SongID (INT): Primary key for the song.
- SongName (VARCHAR(255)): Stores the name of the song.
- AlbumID (INT): Foreign key referencing the Album table.
- Genre (VARCHAR(255)): Stores the genre of the song.
- Length (TIME): Stores the duration of the song.
- Rank (INT): Stores the rank of the song.
- ReleaseDate (DATE): Stores the release date of the song.
- BPM (INT): Beats per minute of the song.
- Features Table:
 - FeatureID (INT): Primary key for the feature.
 - SongID (INT): Foreign key referencing the Song table.
 - FeaturedArtist (VARCHAR(255)): Stores the name of the featured artist.
- Playlist Table:
 - PlaylistID (INT): Primary key for the playlist.
 - PlaylistName (VARCHAR(255)): Stores the name of the playlist.
- Playlist_Songs Table:
 - PlaylistID (INT): Foreign key referencing the Playlist table.
 - SongID (INT): Foreign key referencing the Song table.
 - RankInPlaylist (INT): Stores the rank of the song in the playlist.
- AppUser Table:
 - UserID (INT): Primary key for the user.
 - UserName (VARCHAR(255)): Stores the name of the user.
- User_Playlists Table:
 - UserID (INT): Foreign key referencing the AppUser table.
 - PlaylistID (INT): Foreign key referencing the Playlist table.

Scope/Limitations:

The database could be expanded to include additional features such as user ratings, comments, and social interactions. However, as it currently stands, the database focuses on basic music management functionalities such as organizing playlists, managing user preferences, and storing information about artists, albums, and songs. The database assumes a relatively simple music management system and does not include advanced features such as music recommendation algorithms or streaming capabilities. If the project were to continue and be more all encompassing you could expect more freedom for the user to stream more songs, or for there to be more songs to choose from.